MILO CARROLL

+44 (0) 7715 359626 ◇ London, UK
ucabm49@ucl.ac.uk ◇ linkedin.com/in/milocarroll

RESEARCH SUMMARY

PhD researcher in Artificial Intelligence at UCL working on **motor learning**: how agents learn control from high-dimensional, partially observed sensory streams. My work spans vision-language-action (VLA) policies, latent world modelling (JEPA-style predictive representation learning), diffusion-based policies, and reinforcement learning, applied to humanoid locomotion and robotic manipulation. First-author publications at ICRA and NeurIPS PTA, with further work under review at ICRA and ICLR, alongside applied research at ByteDance Seed and Shell and now owning the control stack at an early-stage robotics company. Core strengths in sequential decision-making under uncertainty, and probabilistic and generative modelling.

TECHNICAL SKILLS

Languages	Python, C++
Frameworks/Tools	PyTorch, Isaac Lab, Isaac Gym, ROS, Linux, Git
Research Areas	Reinforcement Learning, Imitation Learning, Vision-Language-Action (VLA) Models, Latent World Models / Self-Supervised Representation Learning (JEPA), Diffusion and Flow Models, Humanoid Control, Motion Tracking, Sim-to-Real
Robotics Platforms	Unitree A1, Unitree Go2, Unitree G1, Booster T1, Franka
Methods	Experiment design, statistical evaluation, ablation studies, large-scale distributed training, sim-to-real policy transfer

PUBLICATIONS

Start-Controlled Evaluation of JEPA Controllers: Test-Time Search Collapses on Manipulation, and Amortization Needs a Horizon It Should Infer *NeurIPS PTA 2026*
First Author. Identifies weaknesses in current evaluation frameworks for latent world-model controllers and proposes a more robust protocol that accurately reflects observed policy behavior. [OpenReview]

Agile and Versatile Robot Locomotion via Kernel-Based Residual Reinforcement Learning *ICRA 2023*
First Author. Learned MPC motion priors and adapted them with residual RL for robust blind locomotion on the Unitree A1, outperforming the MPC baseline. [IEEE Xplore]

RAI: Recursive Accelerated Imitation for Robot Manipulation *ICRA 2027, Under Review*
First Author. Recursive self-improvement scheme deploying the policy at 2× speed to collect out-of-distribution fast-behavior data, driving success toward 100% at 5× the base cycle rate.

Controllability of Pretrained Visual Representations under Start-Controlled Evaluation on Multi-Object Scenes *ICLR 2027, Under Review*
First Author. New latent world-model representation, raising cube-stacking success from 1% to over 36%.

Efficient Distillation of Deep Thought VLAs: Rectified Depth *CVPR 2027, In Preparation*
Shows the recurrent-depth transformer action head is a contraction, distilling its 32-step converged latent into a single-step generator matching full depth (93.1% vs. 90.5% on LIBERO) at 8.7× lower latency (250 Hz inference).

RESEARCH PROJECTS

SCDP: Learning Humanoid Locomotion from Partial Observations via Mixed-Observation Distillation *arXiv Preprint*
First Author. Diffusion-based humanoid locomotion policy trained with mixed-observation supervision, conditioning on onboard sensor histories while supervising against privileged future state-action trajectories, removing the need for explicit state estimation at deployment. [arXiv:2603.09574]

INDUSTRY EXPERIENCE

Part-Time Applied Research Scientist May 2026 - Present
Cortica (Stealth), London

- Own the low-level control stack for robotic manipulators, from policy training through to on-hardware deployment.

- Built a VR-based teleoperation stack for collecting human demonstration data.
- Designed the pre-training and post-training pipeline for multi-modal robot policies, including force estimation from proprioceptive signals.
- Developed a training protocol for fast-acting policies achieving high success and cycle rates on industrial tasks (see research projects above).
- Supported the founder in VC investment committee meetings and fundraising materials.

AI Scientist Intern Oct 2024 - Oct 2025
ByteDance Seed, London

- Researched motion-tracking policies for humanoid control in dynamic tasks, with a focus on soccer-style behaviors.
- Built motion retargeting pipelines from AMASS/SMPL motion data to humanoid robot platforms including Unitree G1 and Booster T1.
- Developed a kinematic diffusion-based motion generator to produce reference trajectories for low-level motor control.
- Developed and optimized low-level motor policies for smooth and robust motion tracking in simulation.

Co-Founder & CTO Jun 2023 - Sept 2023
Signature Store, London

- Built the technical foundation for an affiliate-commerce platform, secured pre-seed funding, and onboarded initial beta users.

Founder in Residence Apr 2023 - Jun 2023
Entrepreneur First, London

- Explored robotics for logistics and manufacturing, including autonomous multi-machine tending and mobile manipulation, alongside commercialization strategies for generalized robot skills.

Deep Reinforcement Learning AI Resident Sept 2022 - Apr 2023
Shell, London

- Led an RL research project on solar curtailment, investigating control strategies for reducing costs under negative electricity prices while exploiting market inefficiencies with flexible assets.
- Built a simulation environment for German intraday power markets to support policy learning and evaluation.
- Developed an RL-based method for real-option valuation with performance matching state-of-the-art baselines.

Software Engineer Jun 2019 - Sept 2020
Flavrfood, Leeds

- Built MVP and production web applications from scratch, contributing to early company growth and a seed round; developed data collection tools for pricing and availability analysis.

EDUCATION

University College London 2023 - 2027
PhD in Artificial Intelligence, Foundational AI CDT
Supervisor: Alex Li ◇ Autonomous Intelligent Robotics Lab
Thesis area: motor learning, spanning vision-language-action policies, latent world models (JEPA), diffusion policies, and reinforcement learning for control under partial observability.

The University of Edinburgh 2021 - 2022
MSc in Artificial Intelligence, Distinction
Relevant Coursework: Reinforcement Learning, Probabilistic Modelling and Reasoning, Natural Language Processing
Thesis (80%): Agile and Versatile Robot Locomotion via Kernel-Based Residual Reinforcement Learning

Newcastle University / University of Leeds 2018 - 2021
BSc in Computer Science, First Class
Thesis (77%): Reinforcement learning for construction automation (Gazebo / ROS)